

Financial Mathematics Practice Worksheet

Chapter 11.4 – Geometric Series (Complete Set: Tasks 1–20)

Tasks 1–20, sorted from easiest to hardest (difficulty 1/5 to 5/5). Key formulas: a finite geometric series with first term a and quotient $k \neq 1$ has sum $s_n = a + ak + ak^2 + \dots + ak^{n-1} = a \cdot (k^n - 1)/(k - 1)$; when $k = 1$, every term equals a and $s_n = a \cdot n$. An infinite geometric series $a + ak + ak^2 + \dots$ converges if and only if $|k| < 1$, in which case its sum is $a/(1 - k)$; if $|k| \geq 1$ the infinite series diverges and has no finite sum. Tasks 17 and 20 additionally draw on two related results about general (not necessarily geometric) infinite series: convergence requires the terms a_n to tend to 0, though this is necessary but not sufficient (the harmonic series $1 + 1/2 + 1/3 + \dots$ is the standard counterexample), and the p -series rule $\sum 1/n^p$ converges if and only if $p > 1$. All tasks below rely exclusively on these results. Round dollar and quantity amounts to 2 decimal places unless otherwise noted.

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Task 1. Startup Revenue Growth Over Five Years

Difficulty: 1/5

A newly launched e-commerce startup earns \$50 million in revenue this year and expects revenue to grow by 10% annually for each of the next four years, forming a finite geometric series with first term $a = 50$ (million dollars) and quotient $k = 1.10$ over $n = 5$ years.

Statements (True / False):

- a) The expected revenue in year 2 is \$55.00 million.
- b) The expected revenue in year 5 is approximately \$73.21 million.
- c) The total revenue expected over the 5-year period is approximately \$305.26 million.
- d) If revenue had instead remained flat at \$50 million per year for 5 years, the total would have been \$250 million — exactly \$60.00 million less than the actual growth-scenario total.
- e) The total revenue expected over the 5-year period is approximately \$328.86 million.

Solution – Task 1

Given:

- a (year-1 revenue) = \$50 million
- k (annual growth quotient) = 1.10
- $n = 5$ years

Formula(s):

- $s_n = a + ak + ak^2 + \dots + ak^{n-1} = a \cdot (k^n - 1) / (k - 1)$, for $k \neq 1$
- Term in year t ($t = 1, \dots, 5$): $a \cdot k^{t-1}$

Step-by-Step Calculations:

Year-2 revenue = $a \cdot k = 50 \times 1.10 = \55.00 million.

Year-5 revenue = $a \cdot k^4 = 50 \cdot (1.10)^4 = 50 \times 1.4641 = \73.205 million $\rightarrow$ \$73.21 million.

$k^5 = 1.10^5 = 1.61051$.

$s_5 = 50 \cdot (1.61051 - 1) / (1.10 - 1) = 50 \times 0.61051 / 0.10 = 50 \times 6.1051 = \305.255 million $\rightarrow$ \$305.26 million.

Flat-revenue total (0% growth, $k = 1$) = $50 \times 5 = \$250.00$ million.

Difference from actual growth total = $305.255 - 250.00 = \$55.26$ million (not \$60.00 million).

Explanation of Each Statement:

- a) **TRUE.** Year-2 revenue is the year-1 revenue multiplied once by the growth factor: $50 \times 1.10 = \$55.00$ million, matching exactly.
- b) **TRUE.** Counting from year 1 (which is simply a itself) up to year 5, the revenue has been multiplied by 1.10 four separate times — once for each step from year 1 to year 2, year 2 to year 3, and so on up to year 5. So the year-5 revenue equals $a \cdot k^4 = 50 \times 1.4641 = 73.205$, which rounds to \$73.21 million.

c) TRUE. Adding up five terms that each grow by a factor of 1.10 over the previous one gives a total of $s_5 = a \cdot (k^5 - 1) / (k - 1)$. Plugging in $a=50$ and $k=1.10$ gives $k^5=1.61051$, so $s_5 = 50 \times 0.61051 / 0.10 = 50 \times 6.1051 = 305.255$, which rounds to \$305.26 million.

d) FALSE. A flat schedule of \$50 million every year for 5 years does total $50 \times 5 = \$250.00$ million — that part checks out. But the true gap between that flat total and the actual growth total of \$305.255 million is $305.255 - 250.00 = \$55.26$ million, not \$60.00 million; the stated gap overstates the real difference by roughly \$4.74 million.

e) FALSE. Recomputing s_5 carefully with $a=50$, $k=1.10$, $n=5$ gives exactly $50 \times (1.61051 - 1) / 0.10 = \305.255 million, which rounds to about \$305.26 million, as already shown in part (c). The figure of \$328.86 million simply does not match this calculation and is incorrect.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: FALSE

Task 2. Perpetual Profit Stream from a Subscription Box Line

Difficulty: 1.3/5

A subscription box company just earned \$2,000 in profit from its newest product line this month. Because customer renewals are expected to halve every month indefinitely, monthly profits are modeled as an infinite geometric series with first term $a = \$2,000$ and quotient $k = 0.5$.

Statements (True / False):

- a) The infinite profit series converges to a finite sum.
- b) The infinite sum of all future monthly profits is \$4,000.00.
- c) The sum of just the first 4 months' profits is \$3,750.00.
- d) The sum of the first 4 months' profits exceeds the infinite sum.
- e) If the quotient were instead $k = 1.5$, the infinite series would still converge, to a sum of \$-4,000.00.

Solution – Task 2

Given:

- a (month-1 profit) = \$2,000
- k (monthly decay quotient) = 0.5

Formula(s):

- Infinite sum ($|k| < 1$): $a + ak + ak^2 + \dots = a/(1 - k)$
- Finite sum of first n terms: $s_n = a \cdot (1 - k^n)/(1 - k)$

Step-by-Step Calculations:

$|k| = 0.5 < 1$, so the series converges.

Infinite sum = $2,000/(1 - 0.5) = 2,000/0.5 = \$4,000.00$.

First 4 terms: $2,000 + 1,000 + 500 + 250 = \$3,750.00$.

$3,750.00 < 4,000.00$, so the partial sum does not exceed the infinite sum.

If $k = 1.5$, then $|k| = 1.5 \geq 1$, so the series diverges and has no finite sum at all.

Explanation of Each Statement:

- a) **TRUE.** Convergence of an infinite geometric series depends only on the size of the ratio between consecutive terms, not on any other feature of the series. Here each month's profit is exactly half of the previous month's, so $|k| = 0.5$, comfortably below 1 — the running total therefore settles toward a fixed number instead of growing without bound as more months are added.
- b) **TRUE.** When $|k|$ is less than 1, the sum of every term added together forever, $a + ak + ak^2 + \dots$, works out to $a/(1-k)$. Substituting $a=2,000$ and $k=0.5$ gives $2,000/0.5 = \$4,000.00$, the value the cumulative profit approaches as more and more months pass.
- c) **TRUE.** The first four monthly profits are 2,000, then $2,000 \times 0.5 = 1,000$, then $2,000 \times 0.25 = 500$, then $2,000 \times 0.125 = 250$. Adding these directly gives $2,000 + 1,000 + 500 + 250 = \$3,750.00$.

d) FALSE. Every term in this series is a positive dollar amount, so each additional month can only add to the running total, never subtract from it — the partial sum after any finite number of months must therefore sit below the full infinite total. Since \$3,750.00 (first 4 months) is indeed less than \$4,000.00 (all months combined), the first four months' total falls short of the infinite total by \$250.00 rather than exceeding it.

e) FALSE. A ratio of $k=1.5$ has absolute value 1.5, which is greater than 1, so the successive terms of the series (2,000, 3,000, 4,500, 6,750, and so on) keep growing larger and larger instead of shrinking toward zero. A series whose terms grow without bound cannot settle on any single finite total, so it has no sum at all; calling it convergent to \$-4,000.00, or to any specific number, misapplies the whole idea of a limiting sum.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: FALSE

Task 3. Declining Annual Deposits into a Sinking Fund

Difficulty: 1.6/5

An investor commits to an unusual sinking-fund plan: this year's deposit is \$800, and each following year's deposit is 90% of the previous year's deposit, continuing indefinitely. This is an infinite geometric series with $a = \$800$ and $k = 0.90$.

Statements (True / False):

- a) The infinite deposit series converges.
- b) The infinite sum of all deposits is \$8,000.00.
- c) The sum of the first 10 deposits is approximately \$5,210.57.
- d) The first 10 deposits together represent about 65% of the infinite total sum.
- e) If the quotient were instead $k = 1.10$, the series would diverge.

Solution – Task 3

Given:

- a (year-1 deposit) = \$800
- k (annual decay quotient) = 0.90

Formula(s):

- Infinite sum ($|k| < 1$): $a/(1 - k)$
- Finite sum of first n terms: $s_n = a \cdot (1 - k^n)/(1 - k)$

Step-by-Step Calculations:

$|k| = 0.90 < 1$, so the infinite series converges.

Infinite sum = $800/(1 - 0.90) = 800/0.10 = \$8,000.00$.

$0.90^{10} = 0.3486784401$.

$s_{10} = 800 \cdot (1 - 0.3486784401)/0.10 = 800 \times 6.513215599 = \$5,210.57$.

Ratio to infinite sum = $5,210.57/8,000.00 = 0.6513 \approx 65.13\% \approx 65\%$.

If $k = 1.10$, $|k| = 1.10 \geq 1$, so the infinite series diverges by definition.

Explanation of Each Statement:

- a) **TRUE.** Since each year's deposit equals 90% of the prior year's deposit, the ratio between consecutive terms is a constant 0.90, and $|0.90|$ is less than 1. That is exactly the condition needed for the running total of an infinite geometric series to approach a fixed limiting value instead of growing forever.
- b) **TRUE.** For a convergent infinite geometric series, the total of every term added together equals the first term divided by $(1 \text{ minus the ratio})$. With $a = \$800$ and $k = 0.90$, that gives $800/(1 - 0.90) = 800/0.10 = \$8,000.00$ as the amount the cumulative deposits approach in the long run.
- c) **TRUE.** The sum of the first 10 deposits is given by $s_{10} = a \cdot (1 - k^{10})/(1 - k)$. Since 0.90 raised to the 10th power is approximately 0.348678, this becomes $800 \times (1 - 0.348678)/0.10 = 800 \times 6.513216 \approx \$5,210.57$.

d) TRUE. Dividing the 10-year total by the full infinite total gives $5,210.57/8,000.00 \approx 0.6513$, or about 65.1%. Since that rounds to roughly 65%, describing the first 10 deposits as representing about 65% of everything ever deposited is an accurate summary.

e) TRUE. A ratio of $k=1.10$ has absolute value greater than 1, meaning each deposit would be larger than the one before it and would keep growing without any ceiling. A series whose terms increase forever can never settle down to a single finite total, so instead of converging it diverges: the running total simply keeps climbing and never levels off at any particular value.

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: TRUE

Task 4. Lithium Reserve Depletion for a Mining Company

Difficulty: 1.9/5

A mining company estimates its lithium reserves at 18,000,000 tons. Current annual extraction is a constant 300,000 tons per year. Separately, analysts also model a scenario in which extraction instead starts at 300,000 tons and grows by 5% per year for 10 years, a finite geometric series with $a = 300,000$ and $k = 1.05$.

Statements (True / False):

- a) At a constant extraction rate of 300,000 tons/year, the reserves will last exactly 60 years.
- b) If extraction were instead held constant at 500,000 tons/year, the reserves would last 36 years.
- c) Constant-rate extraction over n years is the special $k = 1$ case of the geometric series, where the total simply equals the number of years multiplied by the yearly amount.
- d) If extraction instead grows by 5% per year for 10 years, total extraction over those 10 years is approximately 3,900,000 tons.
- e) The 10-year total under 5% annual growth exceeds the 10-year total under constant 300,000-ton extraction by more than 1,000,000 tons.

Solution – Task 4

Given:

- Reserves = 18,000,000 tons
- Constant extraction $a = 300,000$ tons/year (part a); 500,000 tons/year (part b)
- Growth scenario: $a = 300,000$ tons, $k = 1.05$, $n = 10$ years

Formula(s):

- Years to exhaustion at constant rate: $t = \text{Reserves} / (\text{annual extraction})$
- Constant terms ($k = 1$ case): $s_n = a \cdot n$
- Finite geometric series ($k \neq 1$): $s_n = a \cdot (k^n - 1) / (k - 1)$

Step-by-Step Calculations:

$$t = 18,000,000 / 300,000 = 60 \text{ years.}$$

$$t = 18,000,000 / 500,000 = 36 \text{ years.}$$

Constant extraction of a each year for n years sums to $a \cdot n$ – exactly the $k = 1$ case, where every term in the series is identical.

$$1.05^{10} = 1.628894627.$$

$$s_{10} = 300,000 \cdot (1.628894627 - 1) / 0.05 = 300,000 \times 12.57789254 = 3,773,367.76 \approx 3,773,368 \text{ tons.}$$

$$\text{Constant-rate 10-year total} = 300,000 \times 10 = 3,000,000 \text{ tons.}$$

$$\text{Difference} = 3,773,368 - 3,000,000 = 773,368 \text{ tons, which is less than 1,000,000 tons.}$$

Explanation of Each Statement:

- a) TRUE.** If a fixed 300,000 tons is removed every year from an 18,000,000-ton reserve, the number of years before the reserve reaches zero is simply the total reserve divided by the yearly extraction rate:
 $18,000,000/300,000 = 60$ years.
- b) TRUE.** Using the same reasoning with a faster extraction rate of 500,000 tons per year gives
 $18,000,000/500,000 = 36$ years — a higher annual rate depletes the same fixed reserve proportionally sooner.
- c) TRUE.** When every year's extraction amount is identical (no growth at all), the sequence of yearly amounts is a geometric series whose ratio is exactly 1, meaning every term equals the first term a . Adding together n identical copies of a is just n lots of a , i.e., $a \cdot n$, rather than requiring the general ratio-based formula, which would actually be undefined in this case since it involves dividing by $(k-1)=0$.
- d) FALSE.** If extraction instead starts at 300,000 tons and grows 5% every year for 10 years, the total extracted is $s_{10} = 300,000 \times (1.05^{10} - 1)/0.05$. Since 1.05 to the 10th power is about 1.628895, this works out to $300,000 \times 12.577893 \approx 3,773,368$ tons, noticeably below the 3,900,000 tons claimed.
- e) FALSE.** Comparing the growth scenario ($\approx 3,773,368$ tons over 10 years) to a flat 300,000-tons-per-year scenario over the same span ($300,000 \times 10 = 3,000,000$ tons), the extra amount extracted specifically because of the 5% annual growth is $3,773,368 - 3,000,000 = 773,368$ tons. That is well under 1,000,000 tons, so the claimed excess is too large.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: FALSE

Task 5. Coal Reserves and Declining Output

Difficulty: 2.1/5

A coal mining region has 9,000 million tons (9 billion tons) of estimated reserves. This year's output is 180 million tons, and analysts project output will fall by 3% every year forever, an infinite geometric series with $a = 180$ (million tons) and $k = 0.97$. A second scenario considers a steeper 5% annual decline ($k = 0.95$) starting from the same output.

Statements (True / False):

- a) Output in the second year is 174.6 million tons.
- b) The infinite total extracted over all future years, under the 3% decline, is 6,000 million tons.
- c) The reserves will never be fully exhausted, leaving 3 billion tons as permanently “stranded assets.”
- d) If output instead declined 5% per year from the same 180 million-ton starting point, the total extracted would leave less coal stranded than the 3%-decline case.
- e) Under the 3% decline scenario, cumulative extraction after just the first 20 years alone would already exceed the full infinite-horizon total of 6,000 million tons.

Solution – Task 5

Given:

- Reserves = 9,000 million tons
- a (this year's output) = 180 million tons
- $k = 0.97$ (3% annual decline); alternative $k = 0.95$ (5% annual decline)

Formula(s):

- Infinite sum ($|k| < 1$): $a/(1 - k)$
- Finite sum of first n terms: $s_n = a \cdot (1 - k^n)/(1 - k)$

Step-by-Step Calculations:

Year-2 output = $a \cdot k = 180 \times 0.97 = 174.6$ million tons.

Infinite sum (3% decline) = $180/(1 - 0.97) = 180/0.03 = 6,000$ million tons.

Stranded reserves (3% case) = $9,000 - 6,000 = 3,000$ million tons (3 billion tons).

Infinite sum (5% decline) = $180/(1 - 0.95) = 180/0.05 = 3,600$ million tons.

Stranded reserves (5% case) = $9,000 - 3,600 = 5,400$ million tons – more than the 3,000 million tons stranded under the 3% case.

$0.97^{20} \approx 0.5438$.

$s_{20} = 180 \cdot (1 - 0.5438)/0.03 = 180 \times 15.207 \approx 2,737.3$ million tons, which is far less than 6,000 million tons.

Explanation of Each Statement:

- a) **TRUE.** Year-2 output is the year-1 output multiplied once by the decline factor: $180 \times 0.97 = 174.6$ million tons, matching exactly.

b) TRUE. Because $|0.97|$ is below 1, the infinite series of yearly outputs converges, and its total equals $a/(1-k) = 180/(1-0.97) = 180/0.03 = 6,000$ million tons, the theoretical maximum that could ever be extracted if the 3% annual decline continued indefinitely.

c) TRUE. Comparing the 6,000 million-ton theoretical maximum to the 9,000 million-ton physical reserve shows the reserve can never be fully used up, no matter how long extraction continues, because the running total keeps approaching 6,000 million tons but never reaches or exceeds 9,000 million tons. The difference, $9,000 - 6,000 = 3,000$ million tons, simply remains in the ground indefinitely.

d) FALSE. With a steeper 5% annual decline ($k=0.95$), the theoretical maximum total extraction becomes $180/(1-0.95) = 180/0.05 = 3,600$ million tons, leaving $9,000 - 3,600 = 5,400$ million tons unextracted — more stranded coal than the 3,000 million tons left behind under the milder 3% decline, not less. A steeper decline in output means the total amount ever pulled out is smaller, so more coal is left stranded, not less.

e) FALSE. Summing only the first 20 years under the 3% decline gives $s_{20} = 180 \times (1 - 0.97^{20}) / 0.03$. Since 0.97 to the 20th power is about 0.5438, this comes to roughly $180 \times 15.207 \approx 2,737$ million tons — far below the full 6,000 million-ton theoretical total, since only a fraction of an ever-continuing series has accumulated after just 20 years, not in excess of it.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: FALSE

Task 6. Alternating Correction Payments in a Trading Model

Difficulty: 2.3/5

A quantitative trading model applies a diminishing corrective adjustment after each rebalancing: the first adjustment is \$4,000, and each subsequent adjustment reverses sign and is half the size of the previous one, continuing indefinitely. This is an infinite geometric series with $a = \$4,000$ and a negative quotient $k = -0.5$.

Statements (True / False):

- a) The alternating adjustment series converges to a finite sum.
- b) The infinite sum of all adjustments is approximately \$2,666.67.
- c) The sum of the first 4 adjustments is \$3,000.00.
- d) The series necessarily diverges regardless of its magnitude.
- e) If the quotient were instead exactly $k = -1$, the partial sums would alternate forever between \$4,000 and \$0.

Solution – Task 6

Given:

- a (first adjustment) = \$4,000
- k (sign-reversing decay quotient) = -0.5

Formula(s):

- Infinite sum ($|k| < 1$): $a/(1 - k)$
- Special case $k = -1$: $s_n = a$ when n is odd, $s_n = 0$ when n is even

Step-by-Step Calculations:

$|k| = |-0.5| = 0.5 < 1$, so the series converges.

Infinite sum = $4,000/(1 - (-0.5)) = 4,000/1.5 = \$2,666.67$.

First 4 terms: 4,000, then $4,000 \times (-0.5) = -2,000$, then $4,000 \times 0.25 = 1,000$, then $4,000 \times (-0.125) = -500$.

Sum of first 4 terms = $4,000 - 2,000 + 1,000 - 500 = \$2,500.00$ (not \$3,000.00).

For $k = -1$: $s_n = a$ (odd n) or 0 (even n) – the partial sums oscillate forever and never settle on a limit.

Explanation of Each Statement:

- a) **TRUE.** Convergence depends only on the size of the ratio, $|k|$, not on whether that ratio is positive or negative. Here $k = -0.5$, so $|k| = 0.5$, which is below 1 — the alternating series still settles down to a fixed total even though its terms flip sign at every step.
- b) **TRUE.** The infinite-sum formula $a/(1-k)$ applies the same way regardless of the sign of k , as long as $|k| < 1$. Substituting $a = 4,000$ and $k = -0.5$ gives $4,000/(1 - (-0.5)) = 4,000/1.5 \approx \$2,666.67$.
- c) **FALSE.** The first four adjustments are 4,000, then $4,000 \times (-0.5) = -2,000$, then $4,000 \times 0.25 = 1,000$, then $4,000 \times (-0.125) = -500$. Adding these gives $4,000 - 2,000 + 1,000 - 500 = \$2,500.00$, not \$3,000.00.

d) FALSE. The sign of the ratio has no bearing on whether the series converges; only its magnitude matters. A negative ratio simply means the terms alternate between adding to and subtracting from the running total, and as long as $|k|$ stays below 1, those swings shrink over time and the partial sums settle down — exactly what happens here with $|k|=0.5$, so the series converges rather than diverging.

e) TRUE. With $k=-1$, each new term exactly cancels the one before it: starting from a , the running total becomes a , then 0, then a , then 0, and so on forever, oscillating between those two values without ever settling on one number. Since a limit requires the running total to approach a single fixed value, and this sequence keeps flipping back and forth instead, it never reaches a limit and the series diverges.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: TRUE

Task 7. Equal vs. Growing Pension Contributions

Difficulty: 2.5/5

A city pension fund receives equal annual contributions of \$12 million for 15 straight years (a degenerate geometric series with $k = 1$). Fund managers also model an alternative in which contributions instead grow by 4% every year from the same \$12 million starting point, a finite geometric series with $a = 12$ (million dollars), $k = 1.04$, and $n = 15$.

Statements (True / False):

- a) This is the special $k = 1$ case of the geometric series, where the total simply equals the number of payments multiplied by the payment amount.
- b) The total contributions over the 15 years of equal \$12 million payments equal \$180.00 million.
- c) Under the 4%-growth alternative, the 15-year total is approximately \$240.11 million.
- d) The 4%-growth 15-year total exceeds the no-growth total by more than \$65.00 million.
- e) Applying the general formula directly to the no-growth case would require dividing by zero.

Solution – Task 7

Given:

- Equal-payment case: $a = \$12$ million/year, $n = 15$ years, $k = 1$
- Growth case: $a = \$12$ million, $k = 1.04$, $n = 15$ years

Formula(s):

- $k = 1$ case: $s_n = a \cdot n$
- $k \neq 1$ case: $s_n = a \cdot (k^n - 1) / (k - 1)$

Step-by-Step Calculations:

Equal payments: $s_{15} = 12 \times 15 = \180.00 million.

$$1.04^{15} \approx 1.800944.$$

$$s_{15} \text{ (growth case)} = 12 \cdot (1.800944 - 1) / 0.04 = 12 \times 20.0236 = \$240.28 \text{ million.}$$

Difference = $240.28 - 180.00 = \$60.28$ million, which is not more than \$65.00 million.

The general formula's denominator is $(k - 1)$; at $k = 1$ this denominator is 0, so the formula is undefined and the $k = 1$ case must instead use $s_n = a \cdot n$.

Explanation of Each Statement:

- a) **TRUE.** When every payment is identical, there is no changing ratio between consecutive terms — or equivalently, the ratio is exactly 1. Adding n copies of the same number a together is simply $n \times a$, a much simpler formula than the ratio-based one needed when the terms actually change size from year to year, and it sidesteps the division-by-zero problem that the general formula would run into at $k=1$.
- b) **TRUE.** With 15 identical annual payments of \$12 million each, the total is just $12 \times 15 = \$180.00$ million; no compounding or ratio calculation is needed since nothing changes from one year to the next.

c) FALSE. Growing the same \$12 million starting payment by 4% every year for 15 years gives $s_{15} = 12 \times (1.04^{15} - 1) / 0.04$. Since 1.04 to the 15th power is about 1.800944, this becomes $12 \times 20.0236 \approx \240.28 million, noticeably higher than the \$240.11 million claimed.

d) FALSE. The actual gap between the 4%-growth total ($\approx \$240.28$ million) and the flat total (\$180.00 million) is $240.28 - 180.00 = \$60.28$ million. That is not more than \$65.00 million, so the claim overstates how much extra the growth scenario contributes.

e) TRUE. The general ratio-based formula for a finite geometric series has $(k-1)$ in its denominator. If k were set to exactly 1 — the flat, no-growth case — that denominator becomes zero, and dividing by zero is undefined, so the formula simply cannot be evaluated as written. That is precisely why a separate, simpler rule — multiplying the constant payment by the number of years — has to be used whenever every term in the series is identical.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: TRUE

Task 8. Structured Settlement: Finite Payments vs. Perpetual Value

Difficulty: 2.7/5

A structured settlement is set up to pay \$15,000 in its first year, with each subsequent payment 88% of the prior one ($a = \$15,000$, $k = 0.88$). One clause pays out for a finite 8 years; a separate hypothetical clause considers what the payments would total if continued forever.

Statements (True / False):

- a) The second payment in the sequence is \$13,200.00.
- b) The total of the 8 finite payments is approximately \$80,045.68.
- c) If the payments continued forever under the same 88% quotient, the infinite total would be \$130,000.00.
- d) The finite 8-payment total represents more than 75% of the infinite total.
- e) If the decline rate were less steep, say $k = 0.95$ instead of 0.88, the infinite total would be smaller than the infinite total at $k = 0.88$.

Solution – Task 8

Given:

- a (year-1 payment) = \$15,000
- k (annual decay quotient) = 0.88
- $n = 8$ years (finite clause)

Formula(s):

- Finite sum: $s_n = a \cdot (1 - k^n) / (1 - k)$
- Infinite sum ($|k| < 1$): $a / (1 - k)$

Step-by-Step Calculations:

$$\text{Payment 2} = a \cdot k = 15,000 \times 0.88 = \$13,200.00.$$

$$0.88^8 \approx 0.359635.$$

$$s_8 = 15,000 \cdot (1 - 0.359635) / 0.12 = 15,000 \times 5.336379 = \$80,045.68.$$

$$\text{Infinite sum} = 15,000 / (1 - 0.88) = 15,000 / 0.12 = \$125,000.00 \text{ (not } \$130,000.00).$$

$$\text{Ratio: } 80,045.68 / 125,000.00 = 0.6404 \approx 64.04\%, \text{ which is not more than 75\%}.$$

$$\text{With } k = 0.95: \text{ infinite sum} = 15,000 / (1 - 0.95) = 15,000 / 0.05 = \$300,000.00, \text{ which is larger, not smaller, than the } \$125,000.00 \text{ obtained at } k = 0.88.$$

Explanation of Each Statement:

- a) **TRUE.** The second payment is the first payment multiplied once by the decay factor: $15,000 \times 0.88 = \$13,200.00$, matching exactly.
- b) **TRUE.** Summing the first 8 payments uses $s_8 = a \cdot (1 - k^8) / (1 - k)$. Since 0.88 to the 8th power is about 0.359635, this becomes $15,000 \times (1 - 0.359635) / 0.12 = 15,000 \times 5.336379 \approx \$80,045.68$.
- c) **FALSE.** Since $|0.88| < 1$, letting the payments continue forever gives an infinite total of $a / (1 - k) = 15,000 / (1 - 0.88) = 15,000 / 0.12 = \$125,000.00$, not \$130,000.00.

- d) FALSE.** Dividing the 8-payment total by the true infinite total gives $80,045.68/125,000.00 \approx 0.6404$, or about 64%. That is below 75%, so the claim that the finite total represents more than 75% of the infinite total is incorrect.
- e) FALSE.** A larger ratio (0.95 instead of 0.88) means each payment shrinks more slowly from one year to the next, which makes the $(1-k)$ denominator in the infinite-sum formula smaller — and a smaller denominator produces a larger result, not a smaller one. Concretely, $15,000/(1-0.95) = 15,000/0.05 = \$300,000.00$, far larger than the \$125,000.00 obtained at $k=0.88$, so the claimed comparison has the direction backwards.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: FALSE

Task 9. Growing Franchise Royalties Over 12 Years

Difficulty: 2.85/5

A franchisor receives a royalty of \$9,000 in the first year from a new location, with royalties projected to grow 8% per year for 12 years — a finite geometric series with $a = \$9,000$, $k = 1.08$, and $n = 12$. Analysts also compare this to a hypothetical flat (0%-growth) royalty stream.

Statements (True / False):

- a) The finite-sum formula still gives a valid, well-defined 12-year total.
- b) The total royalties collected over the 12 years are approximately \$175,000.00.
- c) The infinite-sum formula can still be legitimately applied to this series, yielding a meaningful total value.
- d) The royalty payment in year 12 alone is approximately \$20,715.85.
- e) If royalties instead grew at 0% for 12 years, the 12-year total would be \$108,000.00, which is \$62,794.15 less than the actual 8%-growth total.

Solution – Task 9

Given:

- a (year-1 royalty) = \$9,000
- k (annual growth quotient) = 1.08
- $n = 12$ years

Formula(s):

- Finite sum ($k \neq 1$): $s_n = a \cdot (k^n - 1) / (k - 1)$
- Term in year t : $a \cdot k^{t-1}$
- Flat case ($k = 1$): $s_n = a \cdot n$

Step-by-Step Calculations:

$$1.08^{12} \approx 2.518170.$$

$$s_{12} = 9,000 \cdot (2.518170 - 1) / 0.08 = 9,000 \times 18.977128 = \$170,794.15.$$

$$1.08^{11} = 2.518170 / 1.08 \approx 2.331639.$$

$$\text{Year-12 payment} = 9,000 \times 2.331639 = \$20,984.75.$$

$$\text{Flat total} = 9,000 \times 12 = \$108,000.00.$$

$$\text{Difference} = 170,794.15 - 108,000.00 = \$62,794.15.$$

Explanation of Each Statement:

a) TRUE. The finite-sum formula $s_n = a \cdot (k^n - 1) / (k - 1)$ only breaks down when its denominator, $k - 1$, equals zero, which happens exactly when $k = 1$. As long as k is anything other than 1 — including values greater than 1, such as 1.08 here — the formula is perfectly well defined and produces a valid total for any fixed, finite number of terms; the separate requirement that $|k| < 1$ only matters when trying to add up infinitely many terms, not a specific finite count of them.

b) FALSE. Computing $s_{12} = 9,000 \times (1.08^{12} - 1) / 0.08$ requires 1.08 raised to the 12th power, which is about 2.518170. That gives $9,000 \times (2.518170 - 1) / 0.08 = 9,000 \times 18.977128 \approx \$170,794.15$, not \$175,000.00.

c) FALSE. The infinite-sum formula $a/(1-k)$ is only meaningful when the terms shrink toward zero as more are added, which requires $|k|$ to be below 1. Here $k=1.08$ is greater than 1, so each successive royalty payment is larger than the last, growing without bound rather than shrinking — there is no finite total for the series to approach, so plugging these numbers into the infinite-sum formula produces a number with no real meaning, whether positive or negative.

d) FALSE. The 12th year's royalty equals the first payment multiplied by the ratio raised to the 11th power, since the first payment itself already corresponds to zero multiplications. Since 1.08 to the 11th power is about 2.331639, that gives $9,000 \times 2.331639 \approx \$20,984.75$, not \$20,715.85.

e) TRUE. Twelve flat annual payments of \$9,000 with no growth at all would total $9,000 \times 12 = \$108,000.00$. Comparing that to the actual 8%-growth total of about \$170,794.15 gives a difference of $170,794.15 - 108,000.00 = \$62,794.15$, confirming the stated gap exactly.

Final Answers — a: TRUE b: FALSE c: FALSE d: FALSE e: TRUE

Task 10. Tech Company Valuation: Finite Growth Plus Terminal Perpetuity

Difficulty: 3/5

A tech company projects revenue of \$4 million this year, growing 20% per year for 6 years (a finite geometric series, $a = 4$ million, $k = 1.20$, $n = 6$). Investors then treat year-6 revenue as the first payment of a separate terminal perpetuity that declines 15% per year forever ($k = 0.85$), used to estimate value beyond year 6.

Statements (True / False):

- a) The 6-year finite sum of projected revenues is approximately \$39.72 million.
- b) Year-6 revenue alone is approximately \$9.95 million.
- c) The finite 6-year series has no valid sum via the finite-sum formula.
- d) Treating year-6 revenue as the first term of a new perpetuity with quotient $k = 0.85$, the terminal value is approximately \$66.36 million.
- e) Combining the 6-year finite total with the terminal perpetuity value gives a combined projected value of less than \$100 million.

Solution – Task 10

Given:

- Finite phase: $a = \$4$ million, $k = 1.20$, $n = 6$ years
- Terminal phase: first term = year-6 revenue, quotient $k = 0.85$, infinite horizon

Formula(s):

- Finite sum ($k \neq 1$): $s_n = a \cdot (k^n - 1) / (k - 1)$
- Term in year t : $a \cdot k^{t-1}$
- Infinite sum ($|k| < 1$): $a / (1 - k)$

Step-by-Step Calculations:

$$1.20^6 = 2.985984.$$

$$s_6 = 4 \cdot (2.985984 - 1) / 0.20 = 4 \times 9.92992 = \$39.72 \text{ million.}$$

$$1.20^5 = 2.48832; \text{ year-6 revenue} = 4 \times 2.48832 = \$9.95 \text{ million.}$$

$$\text{Terminal value} = 9.95328 / (1 - 0.85) = 9.95328 / 0.15 = \$66.36 \text{ million.}$$

$$\text{Combined value} = 39.72 + 66.36 = \$106.08 \text{ million, which is not less than \$100 million.}$$

Explanation of Each Statement:

- a) **TRUE.** Summing 6 years of revenue that grows 20% each year uses $s_6 = 4 \times (1.20^6 - 1) / 0.20$. Since 1.20 to the 6th power is 2.985984, this becomes $4 \times 1.985984 / 0.20 = 4 \times 9.92992 \approx \39.72 million.
- b) **TRUE.** Year-6 revenue equals the starting revenue multiplied by the growth factor raised to the 5th power, since year 1 itself already corresponds to zero multiplications: $4 \times 1.20^5 = 4 \times 2.48832 \approx \9.95 million.
- c) **FALSE.** The finite geometric-series formula only fails to apply when the ratio equals exactly 1, since that would divide by zero; it works perfectly well for any other ratio, whether smaller or larger than 1, as long as only a fixed, finite number of terms are being added. Here $k=1.20$ is simply the growth factor over a fixed 6-year window, and the

formula handles it without any issue — the restriction that $|k|$ be below 1 only applies when summing infinitely many terms, not a finite set of them.

d) TRUE. Treating the \$9.95328 million of year-6 revenue as the starting payment of a new, separately declining perpetuity with ratio 0.85, the infinite total of that new stream is $9.95328/(1-0.85) = 9.95328/0.15 \approx \66.36 million.

e) FALSE. Adding the 6-year finite total (\$39.72 million) to the terminal perpetuity value (\$66.36 million) gives $39.72+66.36 = \$106.08$ million, which is above \$100 million, not below it.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 11. Reverse-Engineering a Retailer's First-Month Restocking Cost

Difficulty: 3.1/5

A retailer's monthly restocking cost is expected to grow 15% every month for 6 months due to supply-chain constraints (quotient $k = 1.15$). Accounting projects that the total restocking cost over all 6 months will equal exactly \$58,000. Management wants to know what the first month's cost, a , must have been to produce that total.

Statements (True / False):

- a) If the first month's cost had instead been exactly \$1,000, the second month's cost would be \$1,150.00.
- b) Solving for the first-month cost gives approximately \$6,625.74.
- c) The 6th month's restocking cost is approximately \$13,326.73.
- d) The combined restocking cost for months 4 through 6 is approximately \$37,930.00.
- e) If the same \$58,000 total had instead been spread evenly, that flat monthly figure would exceed the actual first-month cost of \$6,625.74 found under 15% growth.

Solution – Task 11

Given:

- s_6 (total 6-month cost) = \$58,000
- k (monthly growth quotient) = 1.15
- $n = 6$ months

Formula(s):

- $s_n = a \cdot (k^n - 1) / (k - 1) \Rightarrow a = s_n \cdot (k - 1) / (k^n - 1)$
- Term in month t : $a \cdot k^{t-1}$

Step-by-Step Calculations:

Hypothetical: if month-1 cost were \$1,000, month-2 cost = $1,000 \times 1.15 = \$1,150.00$.

$$k^6 = 1.15^6 = 2.313060766.$$

$$a = 58,000 \times 0.15 / (2.313060766 - 1) = 8,700 / 1.313060766 = \$6,625.74.$$

$$\text{Month-6 cost} = a \cdot k^5 = 6,625.74 \times 2.011357188 = \$13,326.73.$$

$$\text{Months 4-6 cost} = a \cdot (k^3 + k^4 + k^5) = 6,625.74 \times (1.520875 + 1.749006 + 2.011357) = 6,625.74 \times 5.281239 = \$34,992.12.$$

$$\text{Flat monthly figure} = 58,000 / 6 = \$9,666.67, \text{ which is indeed greater than } \$6,625.74.$$

Explanation of Each Statement:

- a) **TRUE.** Regardless of what the actual first-month cost turns out to be, multiplying any given first-month figure by the 1.15 growth ratio gives the second month's cost; for a hypothetical \$1,000 first month, that is $1,000 \times 1.15 = \$1,150.00$, matching exactly.
- b) **TRUE.** Rearranging the finite-sum formula to solve for the first term gives $a = s_6 \cdot (k - 1) / (k^6 - 1)$. Since 1.15 to the 6th power is about 2.313061, that becomes $a = 58,000 \times 0.15 / 1.313061 = 8,700 / 1.313061 \approx \$6,625.74$.

c) TRUE. The 6th month's cost equals the first month's cost multiplied by the ratio raised to the 5th power, since month 1 itself already corresponds to zero multiplications: $6,625.74 \times 1.15^5 = 6,625.74 \times 2.011357 \approx \$13,326.73$.

d) FALSE. Adding the individual costs for months 4, 5, and 6 — that is, $a \cdot k^3 + a \cdot k^4 + a \cdot k^5$ — gives $6,625.74 \times (1.520875 + 1.749006 + 2.011357) = 6,625.74 \times 5.281239 \approx \$34,992.12$, not \$37,930.00.

e) TRUE. Splitting the same \$58,000 total evenly across 6 months gives $58,000/6 \approx \$9,666.67$ per month. Since the actual first month's cost under 15% growth was only about \$6,625.74, and a growing schedule always starts below its own average (because the larger later months pull the average up), the flat monthly figure is indeed higher than the growth scenario's starting cost.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: TRUE

Task 12. University Endowment: Sustainable Declining Payout

Difficulty: 3.3/5

A university endowment pays out \$500,000 in scholarships this year. To keep the endowment sustainable, the payout is planned to shrink by 2% every year forever ($a = \$500,000$, $k = 0.98$), an infinite geometric series. The board also wants to know the cumulative payout over just the first 10 years, and how a steeper 2%-per-year-larger decline ($k = 0.95$) would compare.

Statements (True / False):

- a) The payout in the second year is \$490,000.00.
- b) The infinite total of all future payouts converges to \$25,000,000.00.
- c) The cumulative payout over just the first 10 years is approximately \$4,800,000.00.
- d) The first 10 years of payouts represent approximately 18% of the full infinite-horizon total.
- e) If the decline were instead steeper, at $k = 0.95$, the infinite total would be \$10,000,000.00, which is more than half of the original \$25,000,000.00 infinite total.

Solution – Task 12

Given:

- a (year-1 payout) = \$500,000
- $k = 0.98$ (primary scenario); alternative $k = 0.95$
- $n = 10$ years (for the finite check)

Formula(s):

- Infinite sum ($|k| < 1$): $a/(1 - k)$
- Finite sum of first n terms: $s_n = a \cdot (1 - k^n)/(1 - k)$

Step-by-Step Calculations:

$$\text{Year-2 payout} = a \cdot k = 500,000 \times 0.98 = \$490,000.00.$$

$$\text{Infinite sum} = 500,000 / (1 - 0.98) = 500,000 / 0.02 = \$25,000,000.00.$$

$$0.98^{10} = 0.817072807.$$

$$s_{10} = 500,000 \cdot (1 - 0.817072807) / 0.02 = 500,000 \times 9.146360 = \$4,573,179.83.$$

$$\text{Ratio} = 4,573,179.83 / 25,000,000.00 = 0.18293 \approx 18.29\% \approx 18\%.$$

$$\text{Infinite sum at } k = 0.95: 500,000 / (1 - 0.95) = 500,000 / 0.05 = \$10,000,000.00.$$

Half of the original infinite sum = $25,000,000.00 / 2 = \$12,500,000.00$; since $10,000,000.00 < 12,500,000.00$, it is NOT more than half.

Explanation of Each Statement:

- a) **TRUE.** Year-2 payout is the year-1 payout multiplied once by the decline factor: $500,000 \times 0.98 = \$490,000.00$, matching exactly.
- b) **TRUE.** Because $|0.98|$ is below 1, letting the payouts continue forever produces a finite total of $a/(1-k) = 500,000/(1-0.98) = 500,000/0.02 = \$25,000,000.00$, the theoretical ceiling on everything ever paid out.

c) **FALSE**. Summing only the first 10 years gives $s_{10} = 500,000 \times (1 - 0.98^{10}) / 0.02$. Since 0.98 to the 10th power is about 0.817073, this becomes $500,000 \times 9.146360 \approx \$4,573,179.83$, not \$4,800,000.00.

d) **TRUE**. Dividing the 10-year total by the full infinite total gives $4,573,179.83 / 25,000,000.00 \approx 0.1829$, or about 18.3% — close enough to describe as roughly 18%.

e) **FALSE**. A steeper 5% annual decline ($k=0.95$) gives an infinite total of $500,000 / (1 - 0.95) = 500,000 / 0.05 = \$10,000,000.00$. Half of the original \$25,000,000.00 infinite total is \$12,500,000.00, and \$10,000,000.00 sits below that halfway point, not above it, so the comparison in the claim has the direction backwards.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 13. Marketing Budget: Solving for the Break-Point Year

Difficulty: 3.5/5

A company's marketing budget starts at \$200,000 this year and grows 12% every year ($a = \$200,000$, $k = 1.12$), a finite geometric series. The CFO wants to know the smallest number of years n after which the cumulative (total) marketing spend first surpasses \$3,000,000.

Statements (True / False):

- a) The marketing budget in year 2 is \$224,000.00.
- b) The cumulative spend after 9 years is approximately \$2,955,131.26, which is still below the \$3,000,000 target.
- c) The cumulative spend after 10 years is approximately \$3,600,000.00.
- d) The smallest n for which cumulative spend surpasses \$3,000,000 is $n = 9$.
- e) If the growth rate were only 8% instead of 12%, the 10-year cumulative spend would be approximately \$2,897,312.49, which would still surpass the \$3,000,000 target within 10 years.

Solution – Task 13

Given:

- a (year-1 spend) = \$200,000
- $k = 1.12$ (primary); alternative $k = 1.08$
- Target cumulative spend = \$3,000,000

Formula(s):

- $s_n = a \cdot (k^n - 1) / (k - 1)$

Step-by-Step Calculations:

$$\text{Year-2 budget} = a \cdot k = 200,000 \times 1.12 = \$224,000.00.$$

$$1.12^9 = 2.773078757; s_9 = 200,000 \cdot (2.773078757 - 1) / 0.12 = 200,000 \times 14.775656 = \$2,955,131.26.$$

$$1.12^{10} = 3.105848289; s_{10} = 200,000 \cdot (3.105848289 - 1) / 0.12 = 200,000 \times 17.548736 = \$3,509,747.01.$$

Since $s_9 \approx \$2,955,131.26 < \$3,000,000$ and $s_{10} \approx \$3,509,747.01 > \$3,000,000$, the smallest n at which the target is surpassed is $n = 10$, not $n = 9$.

$$1.08^{10} = 2.158924997; s_{10}(k=1.08) = 200,000 \cdot (2.158924997 - 1) / 0.08 = 200,000 \times 14.486562 = \$2,897,312.49, \text{ which is below } \$3,000,000.$$

Explanation of Each Statement:

- a) **TRUE.** Year-2 budget is the year-1 budget multiplied once by the growth factor: $200,000 \times 1.12 = \$224,000.00$, matching exactly.
- b) **TRUE.** Summing the first 9 years gives $s_9 = 200,000 \times (1.12^9 - 1) / 0.12$. Since 1.12 to the 9th power is about 2.773079, this becomes $200,000 \times 14.775656 \approx \$2,955,131.26$, which indeed falls short of the \$3,000,000 target.

c) FALSE. Summing the first 10 years instead gives $s_{10} = 200,000 \times (1.12^{10} - 1) / 0.12$. Since 1.12 to the 10th power is about 3.105848, this becomes $200,000 \times 17.548736 \approx \$3,509,747.01$, not $\$3,600,000.00$.

d) FALSE. Since the 9-year total ($\approx \$2,955,131.26$) is still below $\$3,000,000$ but the 10-year total ($\approx \$3,509,747.01$) is above it, the cumulative spend crosses the $\$3,000,000$ threshold sometime during year 10. That means the smallest whole number of years for which the running total exceeds $\$3,000,000$ is $n=10$, not $n=9$.

e) FALSE. With a slower 8% growth rate, the 10-year total is $s_{10} = 200,000 \times (1.08^{10} - 1) / 0.08$. Since 1.08 to the 10th power is about 2.158925, this comes to $200,000 \times 14.486562 \approx \$2,897,312.49$, which is still below $\$3,000,000$ — so under this slower growth rate, the target would not actually be reached within 10 years.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: FALSE

Task 14. Charitable Trust: A Deferred, Declining Grant Stream

Difficulty: 3.7/5

A charitable trust will issue its first grant of \$50,000 starting six years from now, with each subsequent annual grant equal to 96% of the previous grant, continuing forever ($a = \$50,000$, $k = 0.96$). Analysts also want the total of just the first 15 grants, and how a steeper decline would compare.

Statements (True / False):

- a) The second grant in the sequence is \$48,000.00.
- b) The infinite total of all future grants is \$1,250,000.00.
- c) The total of the first 15 grants is approximately \$572,392.03.
- d) The first 15 grants represent less than 40% of the infinite total.
- e) If the decline were steeper, at $k = 0.90$ instead of 0.96, the infinite total would be \$500,000.00, which is less than half of the original \$1,250,000.00 infinite total.

Solution – Task 14

Given:

- a (first grant) = \$50,000
- $k = 0.96$ (primary); alternative $k = 0.90$
- $n = 15$ grants (for the finite check)

Formula(s):

- Infinite sum ($|k| < 1$): $a/(1 - k)$
- Finite sum of first n terms: $s_n = a \cdot (1 - k^n)/(1 - k)$

Step-by-Step Calculations:

$$\text{Grant } 2 = a \cdot k = 50,000 \times 0.96 = \$48,000.00.$$

$$\text{Infinite sum} = 50,000/(1 - 0.96) = 50,000/0.04 = \$1,250,000.00.$$

$$0.96^{15} = 0.542086380.$$

$$s_{15} = 50,000 \cdot (1 - 0.542086380)/0.04 = 50,000 \times 11.447840 = \$572,392.03.$$

$$\text{Ratio} = 572,392.03/1,250,000.00 = 0.45791 \approx 45.79\%, \text{ which is NOT less than } 40\%.$$

$$\text{Infinite sum at } k = 0.90: 50,000/(1 - 0.90) = 50,000/0.10 = \$500,000.00; \text{ half of } \$1,250,000.00 \text{ is } \$625,000.00, \text{ and } \$500,000.00 < \$625,000.00.$$

Explanation of Each Statement:

- a) **TRUE.** The second grant is the first grant multiplied once by the decline factor: $50,000 \times 0.96 = \$48,000.00$, matching exactly.
- b) **TRUE.** Because $|0.96| < 1$, letting the grants continue indefinitely produces a finite total of $a/(1 - k) = 50,000/(1 - 0.96) = 50,000/0.04 = \$1,250,000.00$.
- c) **TRUE.** Summing just the first 15 grants gives $s_{15} = 50,000 \times (1 - 0.96^{15})/0.04$. Since 0.96 to the 15th power is about 0.542086, this becomes $50,000 \times 11.447840 \approx \$572,392.03$.

d) FALSE. Dividing the 15-grant total by the full infinite total gives $572,392.03/1,250,000.00 \approx 0.4579$, or about 45.8% — that is above 40%, not below it, so the claim understates how large a share the first 15 grants actually represent.

e) TRUE. A steeper 10% annual decline ($k=0.90$) gives an infinite total of $50,000/(1-0.90) = 50,000/0.10 = \$500,000.00$. Half of the original \$1,250,000.00 total is \$625,000.00, and \$500,000.00 is indeed below that halfway mark, confirming the comparison.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: TRUE

Task 15. Vineyard Yield: A Quarterly Decline Unit-Conversion Trap

Difficulty: 3.9/5

A vineyard's grape yield is expected to decline by 2% every quarter (not every year) over a 5-year (20-quarter) span. The first quarter's yield is 10,000 lbs, so this is a finite geometric series with $a = 10,000$ lbs, $k = 0.98$, and $n = 20$ quarters.

Statements (True / False):

- a) The second quarter's yield is 9,800.00 lbs.
- b) The total yield over the full 5-year span is approximately 166,196.01 lbs.
- c) Substituting $n = 5$ into the same formula gives 48,039.60 lbs, and this would be the correct 5-year total.
- d) The yield in the 20th quarter alone is approximately 6,812.33 lbs.
- e) If the 2%-per-quarter decline continued forever instead of stopping after 20 quarters, the theoretical infinite total of 500,000.00 lbs would be less than the actual 20-quarter total.

Solution – Task 15

Given:

- a (quarter-1 yield) = 10,000 lbs
- k (quarterly decline quotient) = 0.98
- $n = 20$ quarters (5 years)

Formula(s):

- Finite sum: $s_n = a \cdot (1 - k^n) / (1 - k)$
- Infinite sum ($|k| < 1$): $a / (1 - k)$

Step-by-Step Calculations:

Quarter-2 yield = $a \cdot k = 10,000 \times 0.98 = 9,800.00$ lbs.

$0.98^{20} = 0.667607972$.

$s_{20} = 10,000 \cdot (1 - 0.667607972) / 0.02 = 10,000 \times 16.619601 = 166,196.01$ lbs.

Using the wrong exponent, $n = 5$: $0.98^5 = 0.903920800$; $s_{5, \text{wrong}} = 10,000 \cdot (1 - 0.903921) / 0.02 = 10,000 \times 4.80396 = 48,039.60$ lbs – this is NOT the correct 20-quarter total; it only covers the first 5 quarters (about 1.25 years), not 5 years.

Quarter-20 yield = $a \cdot k^{19} = 10,000 \times 0.681230 = 6,812.33$ lbs.

Infinite (hypothetical) sum = $10,000 / (1 - 0.98) = 10,000 / 0.02 = 500,000.00$ lbs, which is MORE than the 166,196.01 lbs 20-quarter total, not less.

Explanation of Each Statement:

- a) **TRUE.** The second quarter's yield is the first quarter's yield multiplied once by the decline factor: $10,000 \times 0.98 = 9,800.00$ lbs, matching exactly.
- b) **TRUE.** Summing all 20 quarters gives $s_{20} = 10,000 \times (1 - 0.98^{20}) / 0.02$. Since 0.98 to the 20th power is about 0.667608, this becomes $10,000 \times 16.619601 \approx 166,196.01$ lbs.

c) FALSE. Plugging $n=5$ into the same formula instead of $n=20$ does produce 48,039.60 lbs as a number, but that figure only represents the total of the first 5 quarters (about 1.25 years), not the full 5-year, 20-quarter span. Using the wrong exponent gives a technically computable but practically meaningless answer to the actual 5-year question, whose correct total is 166,196.01 lbs.

d) TRUE. The 20th quarter's yield equals the first quarter's yield multiplied by the ratio raised to the 19th power:
 $10,000 \times 0.98^{19} \approx 10,000 \times 0.681230 \approx 6,812.33$ lbs.

e) FALSE. If the 2% quarterly decline continued forever instead of stopping at 20 quarters, the theoretical total would be $10,000/(1-0.98) = 10,000/0.02 = 500,000.00$ lbs. Since every additional quarter beyond the 20th still adds a small positive amount, the infinite total must be at least as large as the 20-quarter partial total, and indeed 500,000.00 lbs is well above 166,196.01 lbs, not below it.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 16. Comparing Two Franchise Territories Over 8 Years

Difficulty: 4.1/5

A franchisor compares two territories over 8 years. Territory A starts at a first-year royalty of \$80,000, growing 6% per year ($a = 80,000$, $k = 1.06$). Territory B starts higher, at \$95,000, but grows more slowly, at only 2% per year ($a = 95,000$, $k = 1.02$).

Statements (True / False):

- a) Territory A's 8-year total royalties are approximately \$791,797.43.
- b) Territory B's 8-year total royalties are approximately \$815,382.06.
- c) Territory A's 8-year total royalties exceed Territory B's 8-year total.
- d) In year 8 alone, Territory A's royalty payment of approximately \$120,290.42 exceeds Territory B's year-8 payment of approximately \$109,125.14.
- e) Territory B's 8-year total exceeds Territory A's 8-year total by more than \$30,000.

Solution – Task 16

Given:

- Territory A: $a = \$80,000$, $k = 1.06$
- Territory B: $a = \$95,000$, $k = 1.02$
- $n = 8$ years for both

Formula(s):

- $s_n = a \cdot (k^n - 1) / (k - 1)$
- Term in year t : $a \cdot k^{t-1}$

Step-by-Step Calculations:

$$1.06^8 = 1.593848075; s_{A,8} = 80,000 \cdot (1.593848075 - 1) / 0.06 = 80,000 \times 9.897468 = \$791,797.43.$$

$$1.02^8 = 1.171659381; s_{B,8} = 95,000 \cdot (1.171659381 - 1) / 0.02 = 95,000 \times 8.582969 = \$815,382.06.$$

Comparing totals: \$791,797.43 (A) < \$815,382.06 (B), so Territory A's total does NOT exceed Territory B's.

$$\text{Year-8 payment A} = 80,000 \cdot 1.06^7 = 80,000 \times 1.503630 = \$120,290.42.$$

$$\text{Year-8 payment B} = 95,000 \cdot 1.02^7 = 95,000 \times 1.148686 = \$109,125.14; \text{ indeed } \$120,290.42 > \$109,125.14.$$

$$\text{Difference (B - A)} = 815,382.06 - 791,797.43 = \$23,584.63, \text{ which is NOT more than } \$30,000.$$

Explanation of Each Statement:

- a) **TRUE.** Summing Territory A's 8 years of royalties gives $s_{A,8} = 80,000 \times (1.06^8 - 1) / 0.06$. Since 1.06 to the 8th power is about 1.593848, this becomes $80,000 \times 9.897468 \approx \$791,797.43$.

b) TRUE. Summing Territory B's 8 years of royalties gives $s_{B,8} = 95,000 \times (1.02^8 - 1) / 0.02$. Since 1.02 to the 8th power is about 1.171659, this becomes $95,000 \times 8.582969 \approx \$815,382.06$.

c) FALSE. Comparing the two 8-year totals directly, \$791,797.43 for Territory A is actually smaller than \$815,382.06 for Territory B — Territory B's higher starting royalty outweighs Territory A's faster growth rate over this particular 8-year window, so Territory A's total does not exceed Territory B's; it falls short of it.

d) TRUE. Territory A's year-8 payment is $80,000 \times 1.06^7 \approx 80,000 \times 1.503630 \approx \$120,290.42$, while Territory B's year-8 payment is $95,000 \times 1.02^7 \approx 95,000 \times 1.148686 \approx \$109,125.14$. By year 8, repeated 6% compounding has pushed Territory A's individual payment above Territory B's, even though Territory A's running 8-year total still lags behind — a genuine subtlety in which the faster-growing stream overtakes the slower one term-by-term before it overtakes it in cumulative total.

e) FALSE. The actual difference between the two totals is $815,382.06 - 791,797.43 = \$23,584.63$, which is below \$30,000, not above it.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 17. Diminishing Marginal Returns from Advertising Spend

Difficulty: 4.3/5

An analyst models the marginal benefit of the n -th batch of advertising spend as $a_n = 5,000/n^p$ dollars, and wants to know, for different values of the exponent p , whether the infinite total marginal benefit $\sum a_n$ converges, using the rule that $\sum 1/n^p$ converges if and only if $p > 1$.

Statements (True / False):

- a) With $p = 1.5$, the marginal benefit of the 4th campaign batch is approximately \$650.00.
- b) With $p = 1.5$, the infinite series converges to a finite total value.
- c) If instead $p = 1$, the series would still converge, just to a larger total than the $p = 1.5$ case.
- d) $a_{100} = \$5.00$, and this alone is enough to guarantee that the series $\sum a_n$ converges.
- e) With $p = 0.5$, the series diverges, even though the individual terms still tend to 0 as $n \rightarrow \infty$.

Solution – Task 17

Given:

- $a_n = 5,000/n^p$
- $p = 1.5$ (primary); $p = 1$ and $p = 0.5$ (comparison scenarios)

Formula(s):

- p -series rule: $\sum_{n=1}^{\infty} 1/n^p$ is convergent $\Leftrightarrow p > 1$
- Necessary (not sufficient) condition for convergence: if $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$

Step-by-Step Calculations:

$$4^{1.5} = 4 \times \sqrt{4} = 4 \times 2 = 8; \quad a_4 = 5,000/8 = \$625.00 \text{ (not \$650.00)}.$$

$p = 1.5 > 1$, so by the p -series rule $\sum 5,000/n^{1.5}$ converges to some finite value.

At $p = 1$, the series $\sum 5,000/n$ is exactly the classic harmonic-series pattern, which can be shown to diverge by grouping terms into blocks that each sum to more than $1/2$ – it does NOT converge to a larger sum; it has no finite sum at all.

$100^{1.5} = 100 \times \sqrt{100} = 100 \times 10 = 1,000$; $a_{100} = 5,000/1,000 = \5.00 – correct value, but terms tending to 0 is necessary, NOT sufficient, for convergence (the harmonic series is the standard counterexample).

$p = 0.5 \leq 1$, so by the p -series rule $\sum 5,000/n^{0.5}$ diverges, even though $a_n = 5,000/\sqrt{n} \rightarrow 0$ as $n \rightarrow \infty$.

Explanation of Each Statement:

- a) **FALSE.** Raising 4 to the power 1.5 means multiplying 4 by its own square root: $4 \times \sqrt{4} = 4 \times 2 = 8$. Dividing the numerator by that gives $5,000/8 = \$625.00$, not \$650.00.
- b) **TRUE.** A series of the form $\sum 1/n^p$ behaves in a very specific way depending on how large p is: when p is bigger than 1, the terms shrink toward zero quickly enough that adding infinitely many of them still produces a finite total, but when p is 1 or smaller, the terms don't shrink fast enough and the running total grows without bound. Since $p=1.5$ here is greater than 1, this series falls into the convergent category and does approach a finite total.

c) FALSE. Setting $p=1$ turns this into the classic series $1 + 1/2 + 1/3 + 1/4 + \dots$, scaled by 5,000. Even though each individual term keeps getting smaller, grouping the terms into blocks — the next 2 terms, then the next 4, then the next 8, and so on — shows that each such block's sum stays above $1/2$ no matter how far out you go, so infinitely many blocks add up to something unbounded. This series does not converge to a larger sum than the $p=1.5$ case; it has no finite sum at all.

d) FALSE. The individual term $a_{100} = 5,000/100^{1.5} = 5,000/1,000 = \5.00 is calculated correctly. However, terms shrinking toward zero only tells you that a series might converge — it does not by itself guarantee convergence. The $p=1$ case in part (c) is the perfect illustration: its terms also shrink toward zero, yet the series still fails to add up to any finite total, so the shrinking-terms observation alone cannot settle the question either way.

e) TRUE. With $p=0.5$, which is not greater than 1, this series falls outside the convergent range and therefore has no finite total, even though the individual terms $5,000/\sqrt{n}$ do get smaller and smaller as n increases. This mirrors the $p=1$ case exactly: shrinking terms are a necessary feature of any convergent series, but on their own they are never sufficient to force convergence.

Final Answers — a: FALSE b: TRUE c: FALSE d: FALSE e: TRUE

Task 18. Solar Farm: Revenue Growth vs. Rising Maintenance Costs

Difficulty: 4.5/5

A solar farm's annual energy revenue starts at \$150,000 and grows 1% per year ($a = 150,000$, $k = 1.01$), while its annual maintenance cost starts at \$120,000 but grows faster, at 3% per year ($a = 120,000$, $k = 1.03$). Both are finite geometric series. Management compares cumulative profit (revenue total minus cost total) over both a 12-year and a 20-year horizon.

Statements (True / False):

- a) The cumulative 12-year revenue is approximately \$1,902,375.45.
- b) The cumulative 12-year maintenance cost is approximately \$1,703,043.55.
- c) The cumulative 12-year profit is approximately \$199,331.90.
- d) In year 12 alone, revenue of approximately \$167,350.25 still exceeds maintenance cost of approximately \$166,108.06, leaving a net positive of about \$1,242.19.
- e) Extending the horizon to 20 years, cumulative profit falls to approximately \$78,405.66, which is smaller than the 12-year cumulative profit of \$199,331.90.

Solution – Task 18

Given:

- Revenue: $a = \$150,000$, $k = 1.01$
- Maintenance cost: $a = \$120,000$, $k = 1.03$
- Horizons: $n = 12$ years and $n = 20$ years

Formula(s):

- $s_n = a \cdot (k^n - 1) / (k - 1)$
- Cumulative profit = $s_{\text{revenue}} - s_{\text{cost}}$
- Term in year t : $a \cdot k^{t-1}$

Step-by-Step Calculations:

$$1.01^{12} = 1.126825030; s_{\text{rev},12} = 150,000 \cdot (1.126825030 - 1) / 0.01 = 150,000 \times 12.682503 = \$1,902,375.45.$$

$$1.03^{12} = 1.425760887; s_{\text{cost},12} = 120,000 \cdot (1.425760887 - 1) / 0.03 = 120,000 \times 14.192030 = \$1,703,043.55.$$

$$\text{Cumulative 12-year profit} = 1,902,375.45 - 1,703,043.55 = \$199,331.90.$$

$$\text{Year-12 revenue} = 150,000 \cdot 1.01^{11} = 150,000 \times 1.115668 = \$167,350.25. \text{ Year-12 cost} = 120,000 \cdot 1.03^{11} = 120,000 \times 1.384234 = \$166,108.06. \text{ Net} = 167,350.25 - 166,108.06 = \$1,242.19.$$

$$1.01^{20} = 1.220190040; s_{\text{rev},20} = 150,000 \cdot (0.220190040) / 0.01 = \$3,302,850.60. 1.03^{20} = 1.806111235; s_{\text{cost},20} = 120,000 \cdot (0.806111235) / 0.03 = \$3,224,444.94. \text{ Profit}_{20} = 3,302,850.60 - 3,224,444.94 = \$78,405.66, \text{ which is indeed less than the 12-year profit of } \$199,331.90.$$

Explanation of Each Statement:

a) TRUE. Summing 12 years of revenue growing 1% annually gives $s_{\text{rev},12} = 150,000 \times (1.01^{12} - 1) / 0.01$. Since 1.01 to the 12th power is about 1.126825, this becomes $150,000 \times 12.682503 \approx \$1,902,375.45$.

b) TRUE. Summing 12 years of maintenance cost growing 3% annually gives $s_{\text{cost},12} = 120,000 \times (1.03^{12} - 1) / 0.03$. Since 1.03 to the 12th power is about 1.425761, this becomes $120,000 \times 14.192030 \approx \$1,703,043.55$.

c) TRUE. Subtracting the cumulative cost from the cumulative revenue gives $1,902,375.45 - 1,703,043.55 = \$199,331.90$ as the total profit accumulated over the 12-year window.

d) TRUE. In year 12 alone, revenue is $150,000 \times 1.01^{11} \approx 150,000 \times 1.115668 \approx \$167,350.25$, while maintenance cost is $120,000 \times 1.03^{11} \approx 120,000 \times 1.384234 \approx \$166,108.06$. Subtracting gives a net of $167,350.25 - 166,108.06 \approx \$1,242.19$ — revenue is still ahead, but only barely, showing how much the gap between the two streams has narrowed by this point.

e) TRUE. Extending the same calculations to 20 years gives a cumulative revenue of about \$3,302,850.60 and a cumulative cost of about \$3,224,444.94, for a 20-year profit of roughly \$78,405.66, smaller than the 12-year profit of \$199,331.90. This happens because maintenance cost compounds at the faster 3% rate while revenue compounds at only 1%; over a longer horizon the faster-growing cost stream eats into more and more of each year's revenue, so even though annual revenue is still ahead of annual cost as late as year 12, the cumulative profit advantage built up in the earlier years gradually erodes as the cost curve catches up.

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: TRUE

Task 19. Private Equity Fund: Recession vs. Recovery Cash-Flow Scenarios

Difficulty: 4.7/5

A private equity fund's portfolio company currently generates \$2,400,000 in annual free cash flow. Under a “recession” scenario, cash flow is expected to shrink 6% every year indefinitely ($a = 2,400,000$, $k = 0.94$). Under a “recovery” scenario, cash flow instead grows 6% per year for the next 7 years before the fund exits ($a = 2,400,000$, $k = 1.06$, $n = 7$).

Statements (True / False):

- a) Under the recession scenario, the infinite cumulative cash flow converges to \$40,000,000.00.
- b) Under the same recession scenario, the cumulative cash flow over just the first 15 years is approximately \$22,000,000.00.
- c) The 15-year recession total represents approximately 75% of the full infinite-horizon recession total.
- d) Under the recovery scenario, the cumulative cash flow over the 7-year period, approximately \$20,145,210.36, exceeds the full infinite-horizon recession total of \$40,000,000.00.
- e) In year 7 of the recovery scenario alone, cash flow is approximately \$2,900,000.00.

Solution – Task 19

Given:

- a (current annual free cash flow) = \$2,400,000
- Recession: $k = 0.94$ (infinite horizon)
- Recovery: $k = 1.06$, $n = 7$ years (finite horizon)

Formula(s):

- Infinite sum ($|k| < 1$): $a/(1 - k)$
- Finite sum of first n terms: $s_n = a \cdot (1 - k^n)/(1 - k)$ or $a \cdot (k^n - 1)/(k - 1)$
- Term in year t : $a \cdot k^{t-1}$

Step-by-Step Calculations:

Infinite recession sum = $2,400,000/(1 - 0.94) = 2,400,000/0.06 = \$40,000,000.00$.
 $0.94^{15} = 0.395291799$; $s_{15} = 2,400,000 \cdot (1 - 0.395291799)/0.06 = 2,400,000 \times 10.078470 = \$24,188,328.05$ (not \$22,000,000.00).

Ratio = $24,188,328.05/40,000,000.00 = 0.60471 \approx 60.47\%$ (not 75%).

$1.06^7 = 1.503630259$; $s_7 = 2,400,000 \cdot (1.503630259 - 1)/0.06 = 2,400,000 \times 8.393838 = \$20,145,210.36$; this is LESS than the infinite recession total of \$40,000,000.00, not more.

Year-7 recovery cash flow = $2,400,000 \cdot 1.06^6 = 2,400,000 \times 1.418519 = \$3,404,445.87$ (not \$2,900,000.00).

Explanation of Each Statement:

- a) **TRUE**. Since the ratio in the recession scenario is 0.94, and $|0.94|$ is below 1, letting the shrinking cash flows continue forever produces a finite total of $a/(1 - k) = 2,400,000/(1 - 0.94) = 2,400,000/0.06 = \$40,000,000.00$.

b) FALSE. Summing just the first 15 years of the recession scenario gives $s_{15} = 2,400,000 \times (1 - 0.94^{15}) / 0.06$. Since 0.94 to the 15th power is about 0.395292 , this becomes $2,400,000 \times 10.078470 \approx \$24,188,328.05$, not $\$22,000,000.00$.

c) FALSE. Dividing the 15-year total by the full infinite total gives $24,188,328.05 / 40,000,000.00 \approx 0.6047$, or about 60%, not 75%.

d) FALSE. Summing the 7-year recovery scenario gives $s_7 = 2,400,000 \times (1.06^7 - 1) / 0.06$. Since 1.06 to the 7th power is about 1.503630 , this becomes $2,400,000 \times 8.393838 \approx \$20,145,210.36$, well below the $\$40,000,000.00$ infinite recession total, not above it.

e) FALSE. The 7th year of the recovery scenario alone is $2,400,000 \times 1.06^6 \approx 2,400,000 \times 1.418519 \approx \$3,404,445.87$, not $\$2,900,000.00$.

Final Answers — a: TRUE b: FALSE c: FALSE d: FALSE e: FALSE

Task 20. Capstone: A Three-Tranche Loan Portfolio and a Convergence Trap

Difficulty: 5/5

A private lender values a portfolio of three loan tranches, valued in nominal (undiscounted) dollars, adding declining and growing streams together just as ordinary fixed amounts. Tranche 1 pays equal annual coupons of \$25,000 for 9 years (the degenerate $k = 1$ case). Tranche 2 pays a growing coupon starting at \$18,000, increasing 7% per year for 9 years ($a = 18,000$, $k = 1.07$). Tranche 3 is a perpetual royalty starting at \$30,000 that declines 8% per year forever ($a = 30,000$, $k = 0.92$). Separately, an analyst proposes valuing a symbolic annual fee stream $f_n = 1,000/n$ (dollars, for $n = 1, 2, 3, \dots$).

Statements (True / False):

- a) Tranche 1 totals exactly \$225,000.00.
- b) Tranche 2 totals approximately \$215,603.80.
- c) Tranche 3, valued as an infinite declining perpetuity, totals \$375,000.00, making it the single largest of the three tranches.
- d) Tranche 3 must be excluded from any combined valuation, meaning the portfolio's correct combined total is just \$440,603.80.
- e) $f_{100} = \$10.00$, and this alone is sufficient to guarantee that the fee stream converges to a finite total.

Solution – Task 20

Given:

- Tranche 1: $a = \$25,000/\text{year}$, $k = 1$, $n = 9$ years
- Tranche 2: $a = \$18,000$, $k = 1.07$, $n = 9$ years
- Tranche 3: $a = \$30,000$, $k = 0.92$, infinite horizon
- Fee stream (separate item): $f_n = 1,000/n$

Formula(s):

- $k = 1$ case: $s_n = a \cdot n$
- Finite sum ($k \neq 1$): $s_n = a \cdot (k^n - 1)/(k - 1)$
- Infinite sum ($|k| < 1$): $a/(1 - k)$
- Necessary (not sufficient) condition for convergence: convergence requires $\lim_{n \rightarrow \infty} a_n = 0$, but this alone does not guarantee convergence, as the harmonic-type counterexample below shows

Step-by-Step Calculations:

Tranche 1: $s = 25,000 \times 9 = \$225,000.00$.

Tranche 2: $1.07^9 = 1.838459212$; $s = 18,000 \cdot (1.838459212 - 1)/0.07 = 18,000 \times 11.977989 = \$215,603.80$.

Tranche 3: $s = 30,000/(1 - 0.92) = 30,000/0.08 = \$375,000.00$; comparing $375,000.00 > 225,000.00 > 215,603.80$, Tranche 3 is indeed the largest.

Combined total (all three) = $225,000.00 + 215,603.80 + 375,000.00 = \$815,603.80$.

This combination is valid: a convergent infinite series ($|k|=0.92<1$) has a specific finite value, which may be added to finite sums just like any other fixed dollar

amount.

Fee stream: $f_{100} = 1,000/100 = \$10.00$ is correct, but this series is the p-series case $p = 1$ (the harmonic series), which is known to diverge – terms shrinking to 0 is necessary but not sufficient for convergence.

Explanation of Each Statement:

a) TRUE. When every one of the 9 annual coupons is identical at \$25,000, there is no changing ratio to apply — the total is simply the number of years multiplied by the constant amount: $25,000 \times 9 = \$225,000.00$.

b) TRUE. Summing 9 years of a coupon that grows 7% annually gives $s = 18,000 \times (1.07^9 - 1)/0.07$. Since 1.07 to the 9th power is about 1.838459, this becomes $18,000 \times 11.977989 \approx \$215,603.80$.

c) TRUE. Since the ratio 0.92 has absolute value below 1, letting this royalty stream continue forever gives a finite total of $30,000/(1-0.92) = 30,000/0.08 = \$375,000.00$. Comparing all three tranches — \$225,000.00, \$215,603.80, and \$375,000.00 — the perpetual royalty stream is indeed the largest of the three, even though it starts with the smallest single payment; over an unlimited horizon, a slowly-declining stream can accumulate more value than either finite stream.

d) FALSE. There is nothing that prevents a convergent infinite series from being added to finite sums — once a series has a specific, finite total, that total behaves just like any other fixed dollar amount and can be combined with other fixed amounts through ordinary addition. Excluding Tranche 3 and reporting only $\$225,000.00 + \$215,603.80 = \$440,603.80$ leaves out \$375,000.00 of value that is just as real and just as calculable as the other two tranches; the correct combined total, including all three, is $225,000.00 + 215,603.80 + 375,000.00 = \$815,603.80$.

e) FALSE. The individual term $f_{100} = 1,000/100 = \$10.00$ is calculated correctly, and it is true that these terms keep shrinking toward zero as n grows. But shrinking toward zero is only a necessary condition for convergence, never a sufficient one — a series can have terms that vanish in the limit and still fail to add up to any finite total. In fact, $f_n = 1,000/n$ is exactly 1,000 times the series $1 + 1/2 + 1/3 + \dots$, which is the standard example of a series whose terms shrink to zero while its running total still grows without any upper bound. The analyst's reasoning is flawed, and no such guarantee of a finite total actually exists here.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: FALSE

This worksheet (Tasks 1–20) relies exclusively on the finite geometric series sum formula, the infinite geometric series sum formula, the convergence condition $|k| < 1$, the necessary-but-not-sufficient zero-limit condition for general series, and the p-series convergence rule.